Networked Pólya Urns and Reinforced Ising Lattices

Quenching, Annealing, and the Dynamics of Social Memory

Lee McCulloch James

The classical Pólya urn and the Ising model come from different intellectual traditions (Eggenberger and Pólya 1923; Ising 1925). One belongs to reinforced probability and path dependence. The other belongs to statistical physics and local alignment. Putting them on the same lattice produces a language for collective behaviour that neither model supplies alone, and that language turns out to fit social, informational, and networked systems with uncomfortable precision.

The motivation is simple. A social agent is influenced by its neighbours and by its own past at the same time. Human and institutional behaviour carries both kinds of pressure at once: historical reinforcement, where prior choices shape future probabilities (Arthur 1989; Pemantle 2007), and local conformity, where neighbouring states pull toward alignment. A model that captures only one of these describes a caricature. The hybrid Pólya-Ising lattice keeps both, and the dynamics that result have memory, locality, reinforcement, and evolving domains all at the same time.

Two interactive implementations accompany this piece.

The first uses contemporaneous neighbour coupling and lives at `https://leelemacjames.github.io/polya-ising-contemporaneous.html`.

The second uses lagged event-memory coupling and lives at `https://leelemacjames.github.io/polya-ising-lattice.html`.

The second grew directly out of limitations that became visible while watching the first one run.

Two lattices, two kinds of social memory

The two implementations accompanying this paper look superficially similar. Both produce evolving domains, local clustering, and large-scale spatial organisation across the lattice. Yet the mechanisms generating those structures differ in a crucial way.

The first model couples each site's reinforced urn directly to the contemporaneous historical fractions of its neighbours. Each neighbour therefore communicates an accumulated identity signal: a compressed summary of its entire path history up to the present moment.

The second model inserts a lagged event-memory field between neighbours and coupling. Neighbours now communicate recent behaviour through an exponentially refreshed memory channel rather than through their full historical fractions. The social field becomes adaptive rather than historically frozen.

The visual distinction between the two regimes is immediate once the lattices evolve.

The contemporaneous model develops heavy historical inertia. Domains become pinned, historical asymmetries persist, and the lattice remembers aggressively. The resulting morphology resembles frozen ideological terrain or entrenched institutional memory.

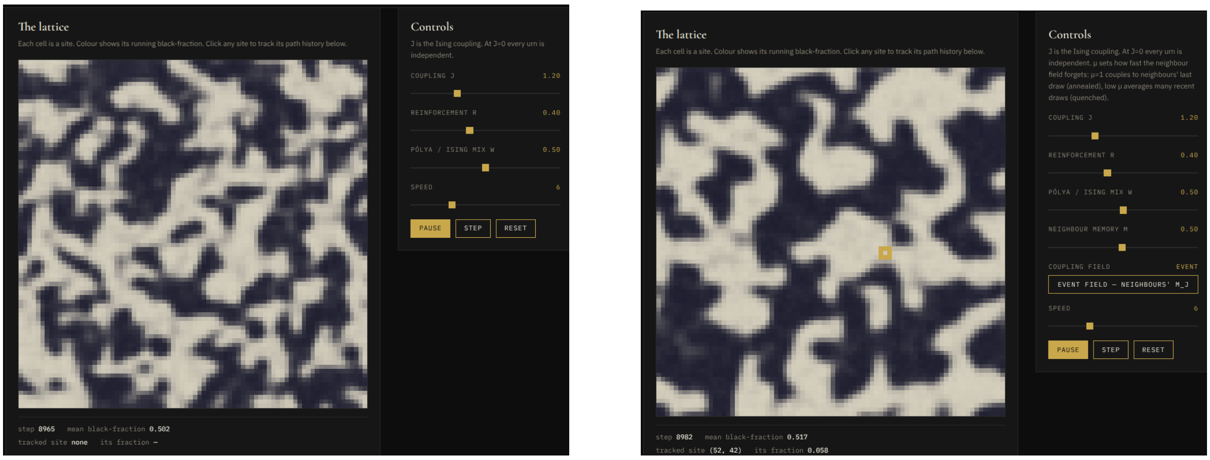

Figure 1: Left: contemporaneous fraction-coupled lattice. Neighbour influence depends directly on reinforced historical fractions, producing highly persistent quenched domains. Right: lagged event-memory lattice. Neighbour influence depends on exponentially refreshed recent behaviour, producing a more annealed and dynamically adaptive morphology.

The lagged model remains more fluid. Boundaries wander, domains reorganise, and local fluctuations continue reshaping the field over long periods. The morphology resembles conversational dynamics, adaptive sentiment fields, or socially revisable consensus.

The distinction reflects two fundamentally different answers to the question:

What exactly does a neighbour communicate?

Does the neighbour communicate accumulated historical identity, or recent behaviour?

The remainder of the paper develops the mathematics and physics behind that distinction.

The Pólya urn that remembers

A Pólya urn begins with coloured balls. The procedure repeats three steps: draw a ball, replace it, then add another ball of the same colour. If B_t and W_t are the black and white counts at time t, the draw probability is

$$P(\text{black at } t) = \frac{B_t}{B_t + W_t},$$

and a black draw updates $B_{t+1} = B_t + 1$. Past draws feed forward into future probability, so the process is reinforced memory in its purest form. Early fluctuations get amplified rather than averaged away, and the system develops historical inertia. The defining property is path dependence: history matters permanently, and it matters in a way that equilibrium thermodynamic systems never reproduce. The urn remembers.

The Ising lattice -the lattice that aligns

The Ising model came out of statistical physics as a model of magnetic alignment. Each site carries a spin $s_i \in \{-1, +1\}$, and neighbouring spins interact through a coupling parameter J:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j.$$

Positive J rewards alignment. At low temperature, neighbouring spins fall into line, domains form, and ferromagnetic order emerges across the lattice. At high temperature, thermal fluctuation dominates, spins decorrelate, and the system stays paramagnetic. The Ising model is therefore a statement about how local interaction produces global coherence, with temperature setting how much coherence survives the noise.

Why the two belong together

The two models reinforce different things. The Pólya urn reinforces an agent's own past; the Ising model reinforces agreement with the neighbourhood. The urn's geometry is temporal, the Ising model's is spatial. The urn produces lock-in, the Ising model produces synchronisation. One asks about path dependence, the other about collective order.

Concept	Pólya urn	Ising model
Primary mechanism	historical reinforcement	neighbour alignment
Memory source	own past	local field
Dominant geometry	temporal	spatial
Typical behaviour	lock-in	synchronisation
Core question	path dependence	collective order

Their combination gives an agent personal memory and social conformity together, which is the sociologically natural arrangement. People and institutions hold onto autobiographical and institutional memory while still responding to the social field around them. The hybrid model is the smallest construction that lets both run at once and lets them interfere.

The contemporaneous coupling model

The first implementation coupled each site's urn directly to its neighbours' current urn fractions. Every site i maintains its reinforced historical fraction b_i/n_i. Neighbour influence enters through a local field built from those fractions,

$$h_i = \sum_{j \sim i} \left(2\frac{b_j}{n_j} - 1 \right),$$

and the draw probability blends the site's own memory with the squashed neighbour field,

$$p_i = (1-w)\frac{b_i}{n_i} + w\,\sigma(Jh_i), \qquad \sigma(x) = \frac{1}{1+e^{-x}}.$$

The weight w sets how much of each pressure the site listens to. The construction does exactly what it was built to do: it blends self-reinforced memory with neighbour conformity. The difficulty is in what the neighbour term is made of. Neighbour influence here depends on accumulated historical fractions, so the social field becomes extremely persistent. A neighbour's b_j/n_j already contains that neighbour's entire path history, and coupling to it means coupling to a frozen historical record rather than to anything the neighbour is currently doing.

Quenching

Quenching, in physics, is rapid freezing before a system has equilibrated. A quenched system preserves whatever structure it had at the moment of the freeze, resists fluctuation, and becomes hard to reconfigure afterward. The contemporaneous model is strongly quenched, and the reason sits in the algebra. The fraction b_i/n_i holds the whole path history of site i, so when neighbour influence reads that fraction it inherits the same long memory. Domain boundaries get pinned. The lattice accumulates historical sediment, layer on layer, and the early configuration keeps asserting itself long after the conditions that produced it have gone.

Quenched dynamics map cleanly onto social structures that remember in this heavy way: institutional memory, entrenched ideology, persistent identity, historical polarisation, reputational inertia. These are all systems whose present is governed by a past they cannot easily revise. The failure mode of the quenched regime is straightforward to name. The system remembers too well.

The lagged event-memory model

The second implementation inserts a layer between the neighbours and the field. Rather than coupling to a neighbour's historical fraction, each site stores an event-memory field $m_i(t)$ that tracks recent behaviour through an exponential moving average:

$$m_i(t+1) = (1-\mu)\, m_i(t) + \mu\, (2X_i(t) - 1),$$

where $X_i(t) \in \{0, 1\}$ is the site's most recent draw. Neighbour coupling now reads this event field instead of the raw fraction,

$$h_i(t) = \sum_{j \sim i} m_j(t),$$

while the draw probability keeps its original shape,

$$p_i = (1-w)\frac{b_i}{n_i} + w\, \sigma(J h_i).$$

The local urn is untouched. Each site still keeps its own full path history in b_i/n_i. The change is confined to the social channel, which now carries a refreshable signal rather than a frozen one.

Annealing

Annealing is the opposite of quenching. In metallurgy it means heating a material so its structure can rearrange, then cooling it slowly so it can settle into a lower-energy configuration. An annealed system stays plastic, explores configurations, allows boundaries to move, and keeps its adaptability. The lagged model anneals the social channel and leaves the local urn memory intact, and that split is the whole point of the construction. Personal history stays rigid. Social influence becomes fluid.

The rate μ governs how fast the social memory refreshes. At $\mu = 1$, neighbour influence depends almost entirely on the most recent draws, the social field fluctuates quickly, and domains stay mobile. At small μ, neighbour influence averages over long stretches of history, the social field turns sluggish, and the behaviour drifts back toward the quasi-quenched regime of the first model. The single slider therefore moves the system along the full axis between the two extremes, which makes the contrast something a reader can watch happen rather than only read about.

What the neighbours actually say

The two models part ways on a single question: what does a neighbour communicate? Under contemporaneous fraction coupling, a neighbour communicates accumulated identity. The social field reflects historical consensus, long-term orientation, entrenched memory. Under lagged event coupling, a neighbour communicates recent behaviour. The social field reflects recent signals, short-term fluctuation, current local momentum. The distinction is the distinction between identity and conversation. One is who the neighbour has been. The other is what the neighbour just did.

Reading the physics as social structure

Social-network researchers meet statistical-physics vocabulary mostly secondhand, so the correspondence is worth stating in full rather than leaving implicit (Castellano et al. 2009).

Physics term	Social-network interpretation
spin alignment	conformity pressure
local field	neighbourhood influence
temperature	randomness / independence
ferromagnetic ordering	consensus formation
paramagnetic disorder	fragmented independence
quenched disorder	frozen historical structure
annealed disorder	fluid adaptive interaction
domain formation	ideological clustering
hysteresis	persistence after stimulus removal
coarsening	growing consensus regions

Where each model applies

The contemporaneous model fits systems that genuinely do remember in the heavy way: long-lived institutions, ideological polarisation, historical identity persistence, reputational lock-in, cultural sedimentation. In all of these the present configuration is held in place by an unrevisable past, and the quenched dynamics reproduce that faithfully.

The lagged event-memory model fits systems whose social channel turns over quickly: rapidly shifting online discourse, financial sentiment, viral contagion, adaptive social influence, trend propagation. Here the relevant signal is what the neighbourhood is doing now, and an annealed field tracks that while a frozen one would lag hopelessly behind.

The systems worth the most attention sit in the middle, holding partial memory and partial adaptability together. Scientific paradigms work this way, and so do political coalitions, institutional trust, social norms, and collective intelligence systems. None of them forgets completely and none of them remembers completely, and the value of the hybrid model is that the intermediate regime is a setting of μ rather than a separate model.

Conclusion

The hybrid Pólya-Ising lattice is a model of social memory under local interaction. It borrows the urn's path dependence and the Ising model's local field, and the combination behaves like neither parent. The contemporaneous version produces highly quenched collective structure, where historical identity dominates the dynamics. The lagged event-memory version anneals the social field, and recent behaviour regains the influence that the frozen version had taken from it.

The distinction matters well beyond physics. Networked systems tend to fail in one of two ways: they forget too fast, or they remember too completely. Collective intelligence usually depends on holding a metastable compromise between the two, plastic enough to take in new information and coherent enough to keep its shape. The two models mark the endpoints of that compromise, and μ is the dial between them.

References

Arthur, W. Brian (1989). "Competing Technologies, Increasing Returns, and Lock-In by Historical Events". In: *The Economic Journal* 99.394, pp. 116–131.

Castellano, Claudio, Santo Fortunato, and Vittorio Loreto (2009). "Statistical Physics of Social Dynamics". In: *Reviews of Modern Physics* 81.2, pp. 591–646.

Eggenberger, F. and George Pólya (1923). ""Uber die Statistik verketteter Vorgänge". In: *Zeitschrift für Angewandte Mathematik und Mechanik* 3.4, pp. 279–289.

Ising, Ernst (1925). "Beitrag zur Theorie des Ferromagnetismus". In: *Zeitschrift für Physik* 31.1, pp. 253–258.

Pemantle, Robin (2007). "A Survey of Random Processes with Reinforcement". In: *Probability Surveys* 4, pp. 1–79.